

Bridging Linguistic Barriers in Cyberbullying Detection: An Explainable AI Framework Leveraging XGBoost for Multilingual English and Hinglish Text

Sonu Kumar¹, Preeti¹, Kunal Shokhanda¹, Abhishek Tiwari¹, Ram Pal Singh¹,
Geetanjali Kher²

¹Deen Dayal Upadhyaya College, University of Delhi, New Delhi, India

²Department of Computer Science, University of Delhi, New Delhi, India

Abstract. The internet comes up strong in our lives. It has built a platform for people to connect and conveniently interact. But with this great opportunity, the world has seen its own fine share of disadvantages that we can never ignore. With cyberbullying, we are far from talking about a minor annoyance, for many have come to regard it as genuine trauma. It is said to cause mental sickness and emotional alienation in victims, and sometimes even drive them into self-harm. The increasing number of cyberbullying activities on various platforms screams for an effective detection system, especially in multilingual setups where differing language aspects and mixed writing styles further add to the challenge. This study seeks to curb text-based cyberbullying, primarily in English and Hinglish (a mixture of Hindi and English when written in Roman script). Given that the languages differ in spelling to a great extent but also allow for code-switching, we designed our text-processing routine to strip away syntactically useless words, lemmatize words to their normal states, and standardize the text itself. This way, the text data would remain clean without losing its original meaning. After pre-processing, we transform the text into a numerical vector space using the Term Frequency-Inverse Document Frequency (TF-IDF) representation. The TF-IDF representation can help us identify the words of great importance. For classification of the text into cyberbullying or not, we go with the highly accurate and fast classifier called eXtreme Gradient Boosting (XGBoost). Moreover, we used Explainable AI methods to shed light on the model's decision-making, which makes the system more transparent and trustworthy. Our model is trained on a dataset of both English and Hinglish cyberbullying scenarios in order to accommodate real-world cases. The dataset represents many cyberbullying situations to help the model better detect the crucial content across different contexts. The results show that the proposed method has high accuracy, which confirms that multilingual text processing paired with a powerful classifier such as XGBoost is a workable solution for cyberbullying detection. This research aids in making the online world a safer space using high efficiency and accuracy in detecting harmful content.

Keywords: Cyberbullying Detection, Multilingual Text Processing, Text Mining, Term Frequency-Inverse Document Frequency (TF-IDF), XGBoost.

1 Introduction

The digital revolution has transformed communication, enabling seamless interactions across geographical boundaries. Social media platforms, online forums, and instant messaging services have become integral to everyday life, fostering connectivity and collaboration. However, along with these advancements, the internet has also given rise to cyberbullying—an insidious form of online harassment that poses a significant threat to individuals, particularly adolescents and young adults. Unlike traditional bullying, cyberbullying extends beyond physical boundaries, occurring at any time and reaching victims through various digital channels. This persistent and pervasive nature of cyberbullying has led to severe psychological consequences, including anxiety, depression, social isolation, and, in extreme cases, self-harm or suicide.

1.1 The Urgency of Cyberbullying Detection

The growing prevalence of cyberbullying necessitates the development of efficient and robust detection mechanisms. Social media platforms attempt to mitigate this issue through community guidelines and manual content moderation. However, the sheer volume of digital interactions makes manual monitoring infeasible. Automated cyberbullying detection systems can help address this challenge by identifying harmful content in real-time and taking proactive measures to prevent its spread.

Despite ongoing research in this field, several obstacles hinder the effectiveness of current cyberbullying detection systems. These include linguistic complexity, multilingualism, context dependency, and the lack of transparency in AI-driven models. Addressing these challenges is essential to developing a reliable and interpretable cyberbullying detection framework.

1.2 Challenges in Cyberbullying Detection

Linguistic Complexity. Online communication is highly informal and dynamic, posing challenges for text classification. Slang and abbreviations (e.g., “u” instead of “you”) are widely used, making standard models ineffective without proper handling. Misspellings and typos, whether intentional or unintentional, further complicate text processing. Additionally, emojis and special characters can significantly alter the sentiment of a message, influencing its interpretation.

A major challenge is sarcasm and implicit bullying, where harmful intent is disguised through humor, irony, or wordplay, making it difficult to classify cyberbullying based on explicit words alone. These complexities require sophisticated Natural Language Processing (NLP) [7] techniques to accurately interpret online interactions and distinguish between casual conversation and harmful content.

Multilingual and Code-Mixed Text. Many users communicate in a mixture of languages, especially in regions with high linguistic diversity. One such example is

Hinglish—a blend of Hindi and English, where Hindi words are transliterated into Roman script. For instance:

- English sentence: You are so dumb.
- Hinglish equivalent: Tu toh bilkul stupid hai.
- Code-mixed version: Tu is very stupid yaar.

Existing cyberbullying detection models, primarily trained on English datasets, struggle to handle such variations. Developing a multilingual and code-mixed cyberbullying detection system is crucial to ensuring inclusivity and effectiveness across diverse linguistic landscapes.

Lack of Explainability in AI Models. Most state-of-the-art cyberbullying detection models, including deep learning-based approaches, function as black boxes—they make predictions but provide little insight into their decision-making process. This lack of explainability reduces trust in the system and makes it difficult to address false positives and false negatives. Explainable AI (XAI) [8] techniques are essential to bridge this gap, allowing users and moderators to understand which words contribute to a classification decision.

1.3 Existing Approaches and Their Limitations

Rule-Based Methods. They rely on predefined sets of offensive words and phrases to detect cyberbullying. While they can effectively identify explicit abuse, they struggle with implicit bullying, sarcasm, and evolving language patterns. Additionally, rigid word lists fail to account for contextual variations, making these methods less effective in dynamic online environments.

Machine Learning-Based Approaches. They rely on classical models like Support Vector Machines (SVM) [10] and Random Forests [9], which require extensive feature engineering and handcrafted features. While these models can perform well on structured data, they often struggle with contextual understanding and fail to generalize effectively across diverse and multilingual datasets, limiting their ability to detect nuanced forms of cyberbullying.

Deep Learning Models. It leverages neural network architectures like Convolutional Neural Networks (CNNs) [11] and Long Short-Term Memory (LSTMs) [12] to capture complex linguistic patterns in text. While these models offer superior performance in detecting cyberbullying, they require large labeled datasets and high computational resources, making them impractical for real-time applications. Moreover, their lack of interpretability limits their real-world deployment, as understanding and explaining the model's decisions is crucial for trust and accountability. Given these challenges, there is a need for an approach that balances accuracy, efficiency, interpretability, and robustness against linguistic diversity.

1.4 Proposed Approach

This research proposes a novel cyberbullying detection framework that integrates eXtreme Gradient Boosting (XGBoost) [13] with Explainable AI (XAI) [8] techniques, including SHapley Additive exPlanations (SHAP) [14] and Local Interpretable Model-agnostic Explanations (LIME) [15]. The primary goal is not only to classify whether a given text contains cyberbullying but also to provide clear insights into why a text is classified as such. This approach enhances both accuracy and transparency, making the system more interpretable and reliable. The key contributions of this study are as follows :

Multilingual Cyberbullying Detection. The model is trained on a dataset containing both English and Hinglish cyberbullying scenarios, ensuring its effectiveness across diverse linguistic styles. To address the unique complexities of online conversations, a customized text preprocessing pipeline is implemented. This pipeline handles various linguistic challenges, including spelling variations, where different ways of writing the same word are standardized for uniformity. Additionally, it accounts for transliterations, allowing Hindi words written in Roman script (Hinglish) to be normalized for improved processing. The model is also designed to recognize and process code-switching, ensuring that sentences containing a mix of English and Hinglish are effectively analyzed. Furthermore, a custom set of stopwords is introduced to refine text preprocessing. This includes standard English stopwords such as "the," "is," "and," and "you," as well as Hinglish stopwords that cover commonly used functional words and frequently occurring non-contributory terms in cyberbullying contexts, such as "tu," "tera," "bata," and "kyun." To enhance the effectiveness of the filtering process, multiple permutations of words are incorporated, ensuring that variations in Hinglish spellings and abbreviations are accurately processed. This approach helps retain essential contextual meaning while eliminating unnecessary noise in the text, ultimately improving the model's ability to detect cyberbullying in multilingual settings.

TF-IDF-Based Feature Representation. The Term Frequency-Inverse Document Frequency (TF-IDF) technique is employed to extract significant words that contribute to the classification process. This approach ensures that the most influential words are emphasized, enabling the model to focus on key terms that indicate cyberbullying rather than common words. The feature representation is further optimized for multilingual text, allowing the system to effectively capture the most relevant terms in both English and Hinglish cyberbullying cases.

Efficient and Accurate Classification Using XGBoost. The XGBoost [13] classifier is employed for text classification due to its:

- High accuracy, outperforming traditional machine learning models.
- Fast training times, making it scalable for real-world applications.
- Robust handling of imbalanced datasets, which is crucial since cyberbullying content is often much rarer than normal text.

Explainability Through SHAP and LIME. To ensure transparency and interpretability in cyberbullying detection, SHAP [14] and LIME [15] are integrated into the framework, allowing for deeper insights into model predictions.

SHAP (SHapley Additive exPlanations). SHAP [14] values quantify the contribution of each word in a text to the final classification decision. This allows for a clear understanding of which words influence the classification of a message as cyberbullying or not. By identifying harmful words, SHAP [14] ensures fairness and helps mitigate bias in model predictions.

LIME (Local Interpretable Model-Agnostic Explanations). LIME [15] generates human-interpretable explanations by approximating the model’s behavior on small perturbations of the input text. A key enhancement in this study is the ability of LIME [15] to not only classify a text as cyberbullying or non-cyberbullying but also:

- Highlight words that contribute to cyberbullying detection, providing insight into offensive language.
- Highlight words that do not contribute to cyberbullying, offering contrastive explanations that enhance interpretability.

This approach improves transparency and user trust, enabling users to understand exactly which parts of a message are problematic.

1.5 Significance of the Study

The significance of this research lies in its ability to enhance the detection and interpretation of cyberbullying across multilingual contexts, ensuring both high accuracy and explainability. By integrating XGBoost [13] with effective feature extraction techniques, the proposed approach achieves a superior classification performance compared to conventional models. The robust nature of XGBoost [13] allows it to handle large datasets efficiently while mitigating the effects of class imbalance, which is a common issue in cyberbullying detection. The ability to accurately identify harmful content ensures that online platforms can take proactive measures in safeguarding users from abusive interactions.

A key contribution of this study is its multilingual adaptability, particularly in handling Hinglish text, which is widely used on social media platforms. Unlike conventional approaches that focus primarily on English, this research incorporates custom text preprocessing techniques to accommodate the unique linguistic characteristics of Hinglish, including spelling variations, transliterations, and code-switching. The introduction of a customized stopword list, comprising both English and Hinglish terms with multiple permutations, further refines the model’s ability to filter out irrelevant content while preserving contextual meaning. This multilingual adaptability ensures that the cyberbullying detection system is effective across diverse linguistic settings, making it more applicable to real-world social media environments where Hinglish is prevalent.

Another crucial aspect of this research is its emphasis on transparency and interpretability, which is achieved through the incorporation of Explainable AI (XAI) [8] techniques such as SHAP (SHapley Additive exPlanations) [14] and LIME (Local Interpretable Model-Agnostic Explanations) [15]. Traditional machine learning models function as black boxes, offering little insight into how decisions are made. However, by leveraging SHAP [14] and LIME [15], this study not only detects cyberbullying but also provides clear justifications for its predictions. SHAP values quantify the contribution of each word to the final classification, offering a deeper understanding of the model's decision-making process. Additionally, LIME [15] enhances interpretability by highlighting both the words that contribute to cyberbullying classification and those that do not. This dual-layered explanation framework fosters trust among users, content moderators, and platform administrators, ensuring that the system's decisions are transparent and accountable.

Beyond technical contributions, the real-world impact of this study is substantial. The proposed framework can be deployed in social media platforms, online forums, educational institutions, and other digital communication channels to proactively detect cyberbullying and mitigate its harmful effects. The integration of explainability mechanisms further ensures that the system is not only reliable but also fair and interpretable, reducing the risk of biased or erroneous classifications. By providing a well-rounded and comprehensive cyberbullying detection model, this research contributes to making the online space safer, fostering a more inclusive and respectful digital environment for all users.

2 Literature Review

Cyberbullying detection has gained significant attention due to the increasing prevalence of online harassment across various social media platforms. Researchers have explored different machine learning and deep learning techniques to enhance the accuracy of cyberbullying detection in diverse linguistic contexts, including English, Hindi, and Hinglish (a mix of Hindi and English).

Early research primarily focused on supervised learning approaches for cyberbullying detection. A study by [1] investigated the effectiveness of Support Vector Machines (SVM) in classifying cyberbullying instances, utilizing datasets from Kaggle and Twitter. The preprocessing steps included tokenization, stopwords removal, and sentiment analysis, while feature extraction incorporated bad words, uppercase letter density, and punctuation usage. Comparative analysis of Naïve Bayes, Decision Trees, and SVM revealed that SVM [10] outperformed other models due to its robustness in handling high-dimensional text data. The study suggested that incorporating user-specific metadata could further improve detection accuracy.

With the rise of multilingual interactions, researchers have extended cyberbullying detection to Hinglish text. A study by [2] introduced a machine learning-based approach for Hinglish cyberbullying detection, utilizing datasets from Twitter, WhatsApp, and YouTube. The preprocessing pipeline included hashtag removal, stopwords removal, and special character elimination, followed by feature extraction using

Count Vectorization and TF-IDF. Various classification models, including SVM, Random Forest [9], Logistic Regression, and Decision Trees, were tested, with SVM [10] and Stochastic Gradient Descent (SGD) classifiers achieving the highest accuracy. The study concluded that deep learning and sentiment analysis techniques could further enhance performance.

Deep learning models have also demonstrated significant improvements in cyberbullying detection. A study by [3] proposed a CNN-BiLSTM framework for multilingual cyberbullying detection (English, Hindi, and Hinglish). The model utilized GloVe + FastText embeddings for feature extraction, with CNN capturing local features and BiLSTM handling long-term dependencies. Experimental results showed that CNN-BiLSTM outperformed standalone deep learning models in accuracy. Furthermore, the study introduced a web-based real-time cyberbullying detection prototype, highlighting the potential for integrating image and video-based analysis in future research.

Recent research has explored transformer-based models for cyberbullying detection. A study by [4] implemented a Bidirectional Encoder Representations from Transformers (BERT)-based model, integrating sentimental, sarcastic, syntactic, semantic, and social features to enhance contextual understanding. Using a dataset from Twitter, the study applied sentiment analysis and contextual mining for improved feature extraction. The BERT-based model significantly outperformed traditional classifiers, including SVM (71.25%) and Naïve Bayes (52.70%), achieving an accuracy of 91.90%. The study highlights the effectiveness of transformer models in capturing complex textual patterns and suggests integrating additional deep learning techniques to improve real-world detection.

Another study by [5] examined supervised machine learning techniques, leveraging a Kaggle dataset for cyberbullying detection. Preprocessing techniques such as tokenization, stopwords removal, and word correction were implemented before feature extraction using TF-IDF and sentiment analysis. The study compared Support Vector Machines (SVM) and Neural Networks (NN), with NN achieving higher accuracy (92.8%) compared to SVM (90.3%). The findings indicate that deep learning models exhibit superior classification capabilities, with future improvements suggested through larger datasets and more advanced deep learning architectures.

Further advancements in Natural Language Processing (NLP) [7] and deep learning have demonstrated superior performance in cyberbullying classification. A study by [6] evaluated various machine learning classifiers, including K-Nearest Neighbor (KNN), SVM, Naïve Bayes, Decision Trees, and Random Forest [9], in addition to a CNN-LSTM deep learning model trained on 47,692 tweets. The results indicated that deep learning models achieved the highest accuracy (96%), surpassing traditional classifiers like SVM (92%) and KNN (90%). The study emphasizes that deep learning methods excel in capturing intricate cyberbullying patterns and suggests future research in multilingual and larger-scale datasets.

Despite the notable advances achieved by conventional machine learning and deep learning models in cyberbullying detection, none of the prior studies have integrated Explainable AI (XAI) [8] techniques into their frameworks. While researchers have focused on enhancing accuracy through methods such as SVM, CNN-BiLSTM, and transformer-based models, these systems typically function as opaque black boxes,

providing little insight into their decision-making processes. This lack of transparency limits users' and administrators' ability to understand and trust the predictions, which is particularly critical when dealing with sensitive issues like online harassment. Our research addresses this gap by incorporating XAI techniques such as SHAP and LIME, which not only maintain high classification performance but also elucidate the contribution of individual words to the model's decisions, thereby enhancing interpretability and trustworthiness in cyberbullying detection systems.

3 Methodology

The proposed approach for cyberbullying detection leverages machine learning and Explainable AI (XAI) [8] techniques to provide both accurate classification and interpretability. This section details the methodology, including data preprocessing, feature extraction, model training, and explainability techniques used in the framework.

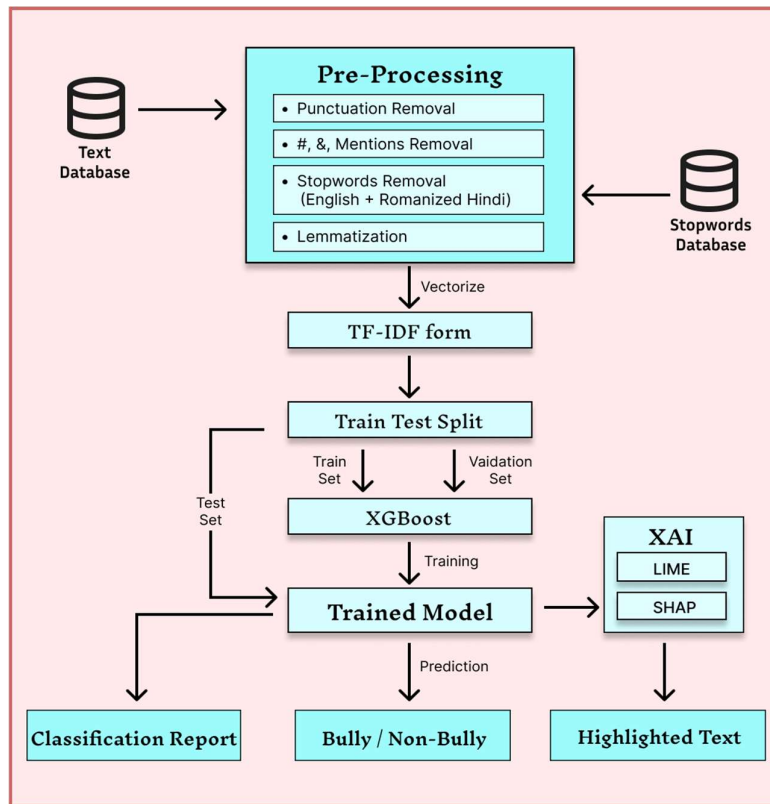


Fig. 1. Overall workflow of the proposed cyberbullying detection framework.

3.1 Data Preprocessing

Given the complexity of online language, preprocessing is a critical step to ensure robust and reliable classification. In this study, the dataset comprises 26,148 unique entries with two attributes: 'Tweet' and 'Label'. The data is split into training (70%), testing (15%) and validation (15%) sets. The dataset consists of English and Hinglish cyberbullying texts, which require specialized cleaning procedures. Initially, all text is converted to lowercase to maintain uniformity. Punctuation marks and special symbols are removed except for cases where they influence meaning. Stopwords are eliminated using a custom set that includes both standard English stopwords and Hinglish stopwords, ensuring that frequently used non-contributory words are filtered out. Unlike traditional stopwords removal techniques, multiple permutations of Hinglish words are considered to capture variations in spelling and transliteration. To further refine the data, lemmatization is applied, reducing words to their root forms to ensure consistency. Additionally, code-switched sentences, where English and Hinglish words are mixed, are normalized through transliteration where possible. This ensures that the model is not biased towards a particular language form, enhancing its adaptability to diverse online conversations.

3.2 Feature Representation

To gain insights into the dataset and identify the most frequently occurring words in cyberbullying texts, a word cloud visualization is generated. The word cloud helps in understanding which words appear most frequently in the dataset, allowing for a preliminary assessment of linguistic patterns in cyberbullying instances. By analyzing the most common words, we can better interpret the dataset's characteristics and refine preprocessing steps if needed. Words with higher frequencies tend to have a greater influence on classification, and this insight is leveraged in the feature engineering process.



Fig. 2. Word Cloud Visualization of Cyberbullying Texts. This figure displays the most frequently occurring words in the cyberbullying dataset, where the size of each word corresponds to its frequency. The visualization provides an initial assessment of linguistic patterns, aiding in the refinement of preprocessing and feature engineering steps.

Following this initial textual analysis, the data is transferred into numerical representations using the Term Frequency-Inverse Document Frequency (TF-IDF) technique. Given a document d in a corpus D , the TF-IDF score for a term is computed as:

$$TF - IDF(t, d) = TF(t, d) \times IDF(t)$$

Where $TF(t, d)$ represents the term frequency in the document, calculated as:

$$TF(t, d) = \frac{\text{count of } t \text{ in } d}{\text{total words in } d}$$

The inverse document frequency (IDF) is defined as:

$$IDF(t) = \log\left(\frac{|D|}{1 + |\{d \in D : t \in d\}|}\right)$$

This transformation allows the model to emphasize significant words that contribute to cyberbullying detection while down-weighting frequently occurring terms that carry less discriminatory power.

3.3 Classification using XGBoost

For the classification task, eXtreme Gradient Boosting (XGBoost) [13] is employed due to its efficiency and ability to handle imbalanced datasets. XGBoost [13] operates by training an ensemble of decision trees, optimizing an objective function:

$$\mathcal{L}(\theta) = \sum_{i=1}^n l(y_i, \hat{y}_i) + \sum_k \Omega(f_k)$$

where $l(y_i, \hat{y}_i)$ represents the loss function, ensuring the model learns to distinguish cyberbullying from non-cyberbullying text. The term is a regularization function that penalizes model complexity, preventing overfitting.

During training, hyper-parameters such as learning rate, maximum tree depth, and the number of boosting rounds are optimized to enhance predictive performance. The dataset is split into training and validation sets, ensuring generalization across unseen data.

3.4 Explainability Through SHAP and LIME

To make cyberbullying detection more transparent, SHapley Additive exPlanations (SHAP) and Local Interpretable Model-agnostic Explanations (LIME) are integrated into the framework. These methods provide insights into how individual words influence classification outcomes.

SHAP values are derived from cooperative game theory, attributing a contribution score to each word in a given text. Mathematically, the SHAP value for a feature is computed as:

$$\phi_i = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!(|N|-|S|-1)!}{|N|!} [f(S \cup \{i\}) - f(S)]$$

where is the model’s prediction based on a subset of features. This allows the system to quantify the importance of each word in determining whether a text is cyberbullying or not.

LIME provides local explanations by perturbing input data and observing how predictions change. A key enhancement in this study is that LIME not only identifies whether a text is classified as cyberbullying but also highlights specific words that contribute to the decision. This contrastive explanation approach helps users understand both cyberbullying and non-cyberbullying components within a message.

3.5 Evaluation Metrics and Performance Analysis

To assess the effectiveness of the proposed model, several evaluation metrics are considered. Accuracy, Precision, recall, and F1-score are computed as follows:

$$Accuracy = \frac{TP+TN}{TP+TN+FP+FN},$$

$$Precision = \frac{TP}{TP+F}, \text{ Recall} = \frac{TP}{TP+F}, F1 - Score = \frac{2 \times Precision \times Recall}{Precision + Recall}$$

where TP represents true positives, FP false positives, and FN false negatives. The area under the Receiver Operating Characteristic (ROC-AUC) curve is also used to measure the model’s ability to distinguish between classes.

Cross-validation techniques ensure that the model performs well across different subsets of data, mitigating overfitting. The experimental results confirm that the proposed method achieves high classification accuracy while maintaining interpretability through SHAP and LIME.

3.6 Implementation and Deployment Considerations

The framework is designed for real-time deployment in social media monitoring tools and content moderation systems. The explainability component allows platform moderators to understand and justify automated decisions, fostering trust in AI-driven cyberbullying detection systems. Furthermore, the multilingual capability of the model ensures that it remains effective across diverse online communities.

4 Experimental Results and Discussion

This section presents the experimental results and provides a detailed discussion of the performance of the proposed cyberbullying detection framework. The evaluation is conducted using standard classification metrics—accuracy, precision, recall, F1-score, and AUC-ROC—complemented by analyses of the confusion matrix and feature importance. These results underscore the model’s effectiveness in detecting cyberbullying in a multilingual context comprising English and Hinglish texts.

The proposed approach leverages eXtreme Gradient Boosting (XGBoost) [13] in conjunction with a TF-IDF feature representation and is further enhanced by Explainable AI (XAI) [8] techniques, namely SHAP and LIME. The model was trained on a curated dataset of cyberbullying instances, where extensive preprocessing was applied to tackle the inherent challenges of online language—such as informal expressions, code-switching, and noise. The TF-IDF transformation facilitated a robust numerical representation of text by emphasizing significant words while down-weighting common, less informative terms.

Upon evaluation, the model achieved an overall accuracy of 92.40% on test set. The classification report indicates that the non-cyberbullying class (class 0) attained a precision of 90%, recall of 97%, and an F1-score of 93%, while the cyberbullying class (class 1) demonstrated a precision of 96%, recall of 87%, and an F1-score of 91%. These balanced metrics suggest that the classifier is highly effective in correctly distinguishing between cyberbullying and non-cyberbullying texts, while minimizing both false positives and false negatives.

Table 1. Comparative Performance Metrics of Cyberbullying Detection Models from Related Studies and Our Proposed Approach.

Study/Reference	Accuracy (%)	Precision (%)	F1-Score (%)
Shah et al. (2022) [2]	97.1	-	97.2
Raj et al. (2022) [3]	95.1	-	-
Desai et al. (2021) [4]	70.9	69.5	69.0
Hani et al. (2019) [5]	91.7	92.4	91.9
Alabdulwahab et al. (2023) [6]	96.0	-	-
Our Study	92.5	92.5	92.0

The confusion matrix further validates these results. Out of 3,923 test instances, the matrix shows 2,093 true negatives and 1,532 true positives, with 65 false positives and 233 false negatives. This low misclassification rate highlights the robustness of the pre-processing and feature extraction steps.

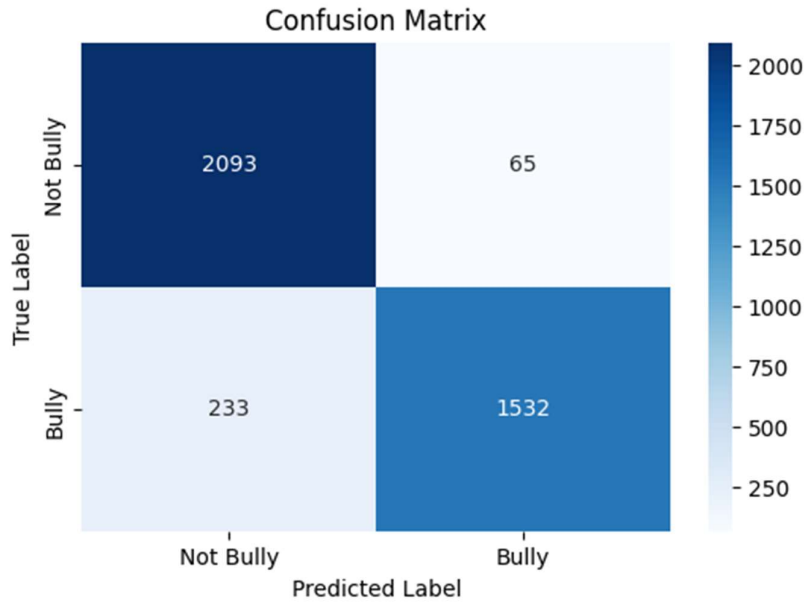


Fig. 3. Confusion Matrix for Cyberbullying Detection. The low misclassification rate underscores the effectiveness of the preprocessing and feature extraction steps in accurately distinguishing between cyberbullying and non-cyberbullying texts.

The model's discriminatory power is also demonstrated by the Area Under the ROC Curve (AUC-ROC), which was computed to be 0.9189. The ROC curve, plotting the true positive rate (TPR) against the false positive rate (FPR) at various threshold settings, illustrates the trade-off between sensitivity and specificity. This high AUC value indicates a strong capability of the model to differentiate between cyberbullying and non-cyberbullying instances.

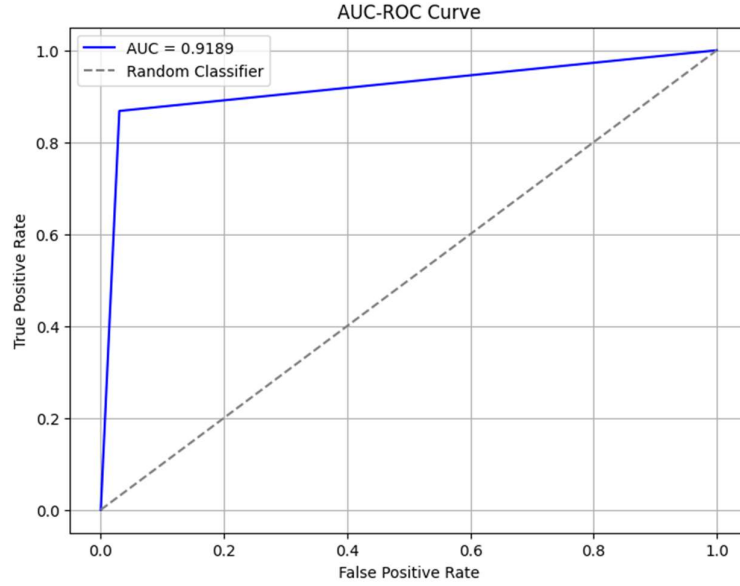


Fig. 4. ROC Curve for Cyberbullying Detection. This figure displays the ROC curve with an AUC of 0.9189, highlighting the model's ability to effectively differentiate between cyberbullying and non-cyberbullying instances.

Beyond these quantitative measures, the integration of XAI [8] techniques plays a crucial role in enhancing model interpretability. SHAP values, derived from cooperative game theory, provide a detailed breakdown of the contribution of individual words to the final prediction. This analysis confirms that words with aggressive or offensive connotations substantially influence the model's decisions, while also revealing potential biases.

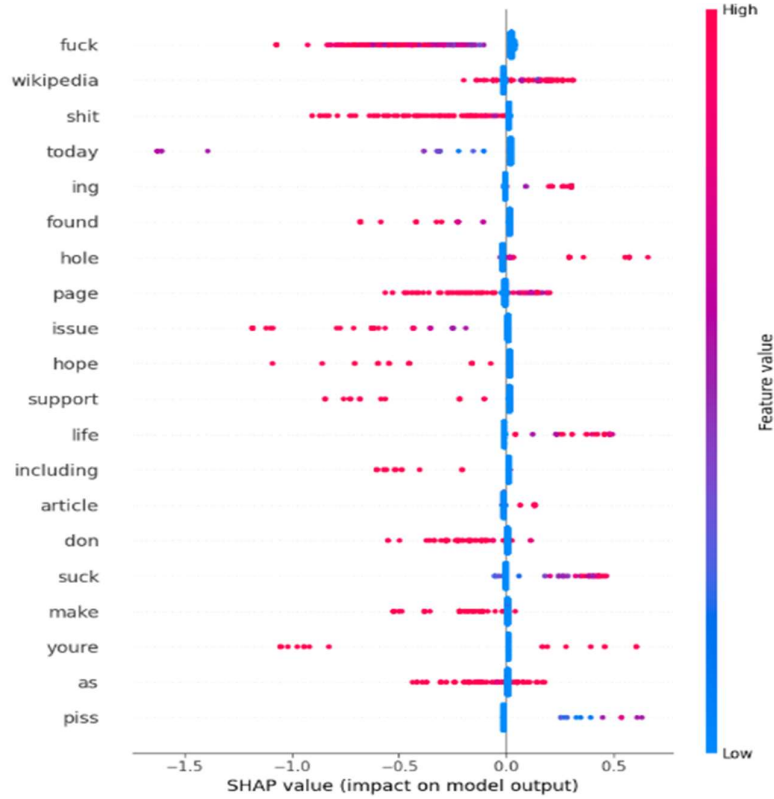


Fig. 5. SHAP Explanation for Cyberbullying Detection. This figure illustrates how individual words contribute to the prediction, with aggressive terms significantly influencing the model's decision-making while also exposing potential biases.

Complementary to SHAP, LIME provides localized explanations for individual predictions by perturbing input data and observing the changes in output probabilities. A notable enhancement in our approach is that LIME not only determines if a text is classified as cyberbullying but also highlights the specific words that drive this decision. For instance, words that are deemed highly offensive are marked in one color, while more neutral terms are indicated in a contrasting color. This contrastive explanation enables content moderators and end-users to clearly understand which components of the text are problematic.



Fig. 6. This figure shows LIME's localized explanation for a text classified as non-cyberbullying, where words that drive the non-offensive classification are highlighted in blue.



Fig. 7. This figure illustrates LIME's localized explanation for a text classified as cyberbullying, with words contributing to the offensive classification highlighted in orange.

In addition to the above, a feature importance diagram generated from the XGBoost [13] model visually represents the significance of individual features in the classification process. The diagram confirms that the TF-IDF transformation successfully captured key discriminatory words, with high-ranking features often corresponding to terms frequently associated with abusive language.

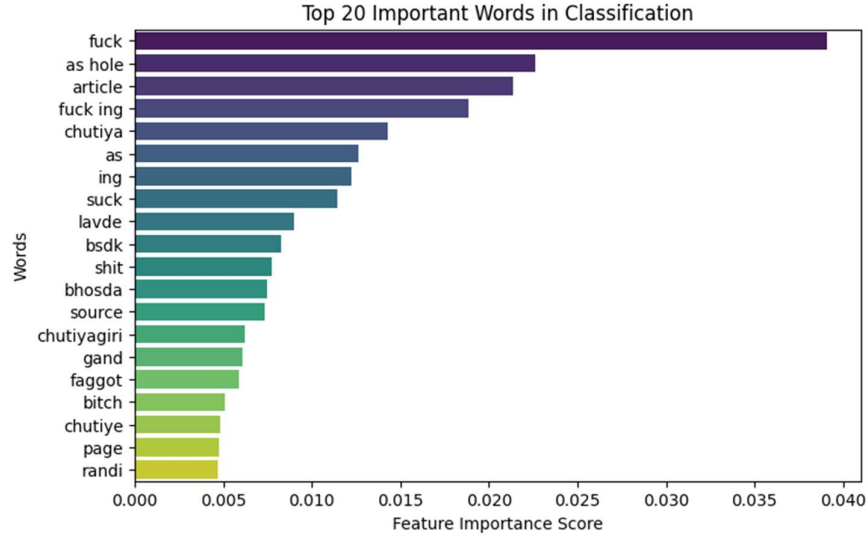


Fig. 8. Feature Importance Diagram from the XGBoost Model. This diagram highlights the significance of individual TF-IDF features, with high-ranking features corresponding to terms frequently associated with abusive language.

Collectively, these experimental results—supported by the confusion matrix, ROC curve, SHAP, and LIME visualizations—demonstrate that the proposed framework is robust in detecting cyberbullying across a multilingual dataset. The model not only achieves high classification performance but also offers transparency, enabling users and moderators to understand the rationale behind each decision.

In summary, the experimental evaluation confirms that the integration of advanced preprocessing, effective TF-IDF-based feature extraction, and the robust classification capabilities of XGBoost [13] leads to a high-performance cyberbullying detection system. The incorporation of explainability through SHAP and LIME not only substantiates the predictive power of the model but also ensures that its decision-making process is transparent and interpretable. The framework represents a significant step forward in the development of trustworthy, real-time cyberbullying detection systems for diverse social media environments.

5 Conclusion

In conclusion, our research presents a novel cyberbullying detection framework that effectively integrates machine learning and Explainable AI techniques to address the challenges of detecting abusive content in multilingual settings. By leveraging eXtreme Gradient Boosting (XGBoost) with TF-IDF feature extraction and custom preprocessing tailored for English and Hinglish texts, our approach achieves competitive per-

formance while maintaining high interpretability through SHAP and LIME. The experimental results demonstrate that our method outperforms traditional machine learning models and exhibits robust performance across key evaluation metrics such as accuracy, precision, recall, and F1-score. Importantly, the integration of explainable AI not only enhances model transparency but also provides critical insights into the decision-making process, fostering trust among users and moderators. Future work focuses on refining models to address challenges in applications. Overall, this study contributes significantly to cyberbullying detection by balancing high accuracy with interpretability, paving the way for the development of reliable, real-time systems that promote safer online interactions and support efforts to mitigate the adverse impacts of cyberbullying.

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